

Problem 1

$$1. D(s) = s^4 + 22s^3 + 10s^2 + 2s + K$$

all coefficients are positive, satisfying the necessary condition
 $\Downarrow$

(all coefficients have same sign)

By using Routh Array (Routh-Hurwitz Criterion)

$$s^4 \quad 1 \quad 10 \quad K$$

$$s^3 \quad 22 \quad 2 \quad 0$$

$$s^2 \quad \frac{109}{11} \quad K \quad 0$$

$$s^1 \quad 2 - \frac{242}{109}K \quad 0$$

$$s^0 \quad K$$

In order for the system to be stable, all terms in the first column of Routh table must be positive (have same signs)

$$2 - \frac{242}{109}K > 0, \quad K > 0$$

$$\underline{0 < K < \frac{109}{121}}$$

$$2. D(s) = s^4 + 20Ks^3 + 5s^2 + (10+K)s + 15$$

$$s^4 \quad 1 \quad 5 \quad 15$$

$$s^3 \quad 20K \quad 10+K \quad 0$$

$$s^2 \quad \frac{99K-10}{20K} \quad 15 \quad 0$$

$$s^1 \quad \left(10+K - \frac{6000K^2}{99K-10}\right)$$

$$s^0 \quad 15$$

$$K > 0, \quad \frac{99K-10}{20} > 0 \quad K > \frac{10}{99}$$

$$10+K - \frac{6000K^2}{99K-10} > 0$$

$$10+K > \frac{6000K^2}{99K-10}$$

$$10+K(99K-10) > 6000K^2$$

$$990K-100+99K^2-10K > 6000K^2$$

$$5901K^2 - 980K + 100 < 0$$

$$\frac{980 - \sqrt{980^2 - 4(5901)(100)}}{2(5901)} < K < \frac{980 + \sqrt{980^2 - 4(5901)(100)}}{2(5901)}$$

$$\underline{K > \frac{10}{99}}$$

$$3. D(s) = s^3 + (k + 0.5)s^2 + 4ks + 50$$

$$s^3 \quad | \quad 4k$$

$$k + 0.5 > 0$$

$$s^2 \quad | \quad (k + 0.5)50$$

$$k > -0.5$$

$$s^1 \quad | \quad \frac{(8k^2 + 4k - 100)}{2k + 1}$$

$$8k^2 + 4k - 100 \geq 0$$

$$s^0 \quad | \quad 50$$

$$2k^2 + k - 25 \geq 0$$



$$\frac{-1 - \sqrt{201}}{4} < k < \frac{-1 + \sqrt{201}}{4} \quad \& \quad k \geq -0.5$$

$\Downarrow$

$$\underline{-\frac{1}{2} < k < \frac{-1 + \sqrt{201}}{4}}$$

$$4. D(s) = s^4 + ks^3 + s^2 + s + 1$$

$$s^4 \quad | \quad 1 \quad 1 \quad 1$$

$$\frac{k-1}{k} > 0, \quad k > 0$$

$$s^3 \quad | \quad k \quad 1 \quad 0$$

$\Downarrow$

$$s^2 \quad | \quad \frac{k-1}{k} \quad 1$$

$$-k^2 + k - 1 > 0$$

$$s^1 \quad | \quad \left(\frac{-k^2 + k - 1}{k-1} \right)$$

$$k > 1$$

$$k^2 - k + 1 < 0$$

$$s^0 \quad | \quad 1$$

There is no k value that satisfy the criteria for this system to be stable

Problem 2

$$M=1, g=10, L=1, I = \frac{ML^2}{3} = \frac{1}{3}$$

$$(1) \quad I\ddot{\theta}_1 = -Mgl\theta_1 + (F_s + F_D)l$$

$$F_s = Kl(\theta_2 - \theta_1)$$

$$F_D = B_S l(\dot{\theta}_2 - \dot{\theta}_1)$$

$$I\ddot{\theta}_2 = 2Fl - Mgl\theta_2 - (F_s + F_D)l$$

$$K=1, B_S=0.01$$

$$I\ddot{\theta}_1 = -Mgl\theta_1 + Kl^2(\theta_2 - \theta_1) + B_S l^2(\dot{\theta}_2 - \dot{\theta}_1)$$

$$I\ddot{\theta}_2 = 2Fl - Mgl\theta_2 - Kl^2(\theta_2 - \theta_1) - B_S l^2(\dot{\theta}_2 - \dot{\theta}_1)$$

Laplace Transform:

$$I[s^2\theta_1(s) - s\theta_1(0) - \dot{\theta}_1(0)] = -Mgl\theta_1(s) + Kl^2\theta_2(s) - Kl^2\theta_1(s) + B_S l^2[s\theta_2 - \theta_2(0)] - B_S l^2[s\theta_1 - \theta_1(0)]$$

$$I[s^2\theta_2 - s\theta_2(0) - \dot{\theta}_2(0)] = \frac{2Fl}{s} - Mgl\theta_2 - Kl^2\theta_2 + Kl^2\theta_1 - B_S l^2[s\theta_2 - \theta_2(0)] + B_S l^2[s\theta_1 - \theta_1(0)]$$

Assuming I.C.s = 0, substitute all known values:

$$\frac{1}{3}s^2\theta_1(s) = -10\theta_1(s) + \theta_2(s) - \theta_1(s) + 0.01s\theta_2(s) - 0.01s\theta_1(s)$$

$$\frac{1}{3}s^2\theta_1(s) + 11\theta_1(s) + 0.01s\theta_1(s) = 0.01s\theta_2(s) + \theta_2(s)$$

$$(\frac{1}{3}s^2 + 11 + 0.01s)\theta_1(s) = (0.01s + 1)\theta_2(s)$$

$$\theta_2(s) = \frac{\frac{1}{3}s^2 + 0.01s + 11}{0.01s + 1} \theta_1(s)$$

$$\frac{1}{3}s^2\theta_2(s) = 2F(s) - 10\theta_2(s) - \theta_2(s) + \theta_1(s) - 0.01s\theta_2(s) + 0.01s\theta_1(s)$$

$$0.01s\theta_2(s) + \frac{1}{3}s^2\theta_2(s) + 11\theta_2(s) = +0.01s\theta_1(s) + \theta_1(s)$$

$$(0.01s + \frac{1}{3}s^2 + 11) \left[\frac{\frac{1}{3}s^2 + 0.01s + 11}{0.01s + 1} \theta_1(s) \right] = 2F(s) + (0.01s + 1)\theta_1(s)$$

$$\frac{\frac{1}{9}s^4 + \frac{1}{180}s^3 + \frac{22}{3}s^2 + \frac{1}{5}s + 120}{0.01s + 1} \theta_1(s) = 2F(s)$$

$$\frac{\theta_1(s)}{F} = \frac{0.02s + 2}{\frac{1}{9}s^4 + \frac{1}{180}s^3 + \frac{22}{3}s^2 + \frac{1}{5}s + 120}$$

zeros: $0.02s + 2 = 0$
 $z = -100$

poles: by using matlab

$$p_{1,2} = -0.03 \pm 5.999i \quad p_{3,4} = \pm 5.4772i$$

(2) Assume $\theta_2(0) = 0$

$$\frac{1}{3} [s^2\theta_1(s) - 0.01s] = -10\theta_1(s) + \theta_2(s) - \theta_1(s) + 0.01s\theta_2(s) - 0.01s\theta_1(s) + 0.001$$

$$\frac{1}{3}s^2\theta_2 = 2(0) - 10\theta_2(s) - \theta_2(s) + \theta_1(s) - 0.01s\theta_2(s) + 0.01s\theta_1(s) - 0.001$$

$$\frac{1}{3}s^2\theta_1(s) - \frac{1}{300}s = (-11 - 0.01s)\theta_1(s) + (0.01s + 1)\theta_2(s) + 0.001$$

$$(\frac{1}{3}s^2 + 11 + 0.01s)\theta_1(s) - \frac{1}{300}s - 0.001 = (0.01s + 1)\theta_2(s)$$

$$\frac{\frac{1}{3}s^2 + 11 + 0.01s}{0.01s + 1} \theta_1(s) - \frac{(\frac{1}{300}s + 0.001)}{0.01s + 1} = \theta_2(s)$$

$$(\frac{1}{3}s^2 + 11 + 0.01s)\Theta_2(s) = (1 + 0.01s)\Theta_1(s) - 0.001$$

$$(\frac{1}{3}s^2 + 11 + 0.01s) \left[\frac{\frac{1}{3}s^2 + 11 + 0.01s}{0.01s + 1} \Theta_1(s) - \left(\frac{\frac{1}{300}s + 0.01}{0.01s + 1} \right) \right] = (1 + 0.01s)\Theta_1(s) - 0.001$$

$$\frac{(\frac{1}{3}s^2 + 11 + 0.01s)^2}{0.01s + 1} \Theta_1(s) - \frac{(\frac{1}{3}s^2 + 0.01s + 11)(\frac{1}{300}s + 0.01)}{0.01s + 1} = (1 + 0.01s)\Theta_1(s) - 0.001$$

$$\frac{(\frac{1}{3}s^2 + 11 + 0.01s)^2 - (1 + 0.01s)^2}{0.01s + 1} \Theta_1(s) = \frac{(\frac{1}{3}s^2 + 0.01s + 11)(\frac{1}{300}s + 0.01)}{0.01s + 1} - 0.001$$

$$\Theta_1(s) = \frac{(\frac{1}{3}s^2 + 0.01s + 11)(\frac{1}{300}s + 0.01) - 0.0000(s - 0.001)}{0.01s + 1} \times \frac{0.01s + 1}{(\frac{1}{3}s^2 + 11 + 0.01s)^2 - (1 + 0.01s)^2}$$

$$\Theta_1(s) = \frac{\frac{s^3}{900} + \frac{101s^2}{30000} + \frac{11027s}{30000} + \frac{101}{1000}}{\frac{s^4}{9} + \frac{s^3}{150} + \frac{22s^2}{3} + \frac{6}{25}s + 122}$$

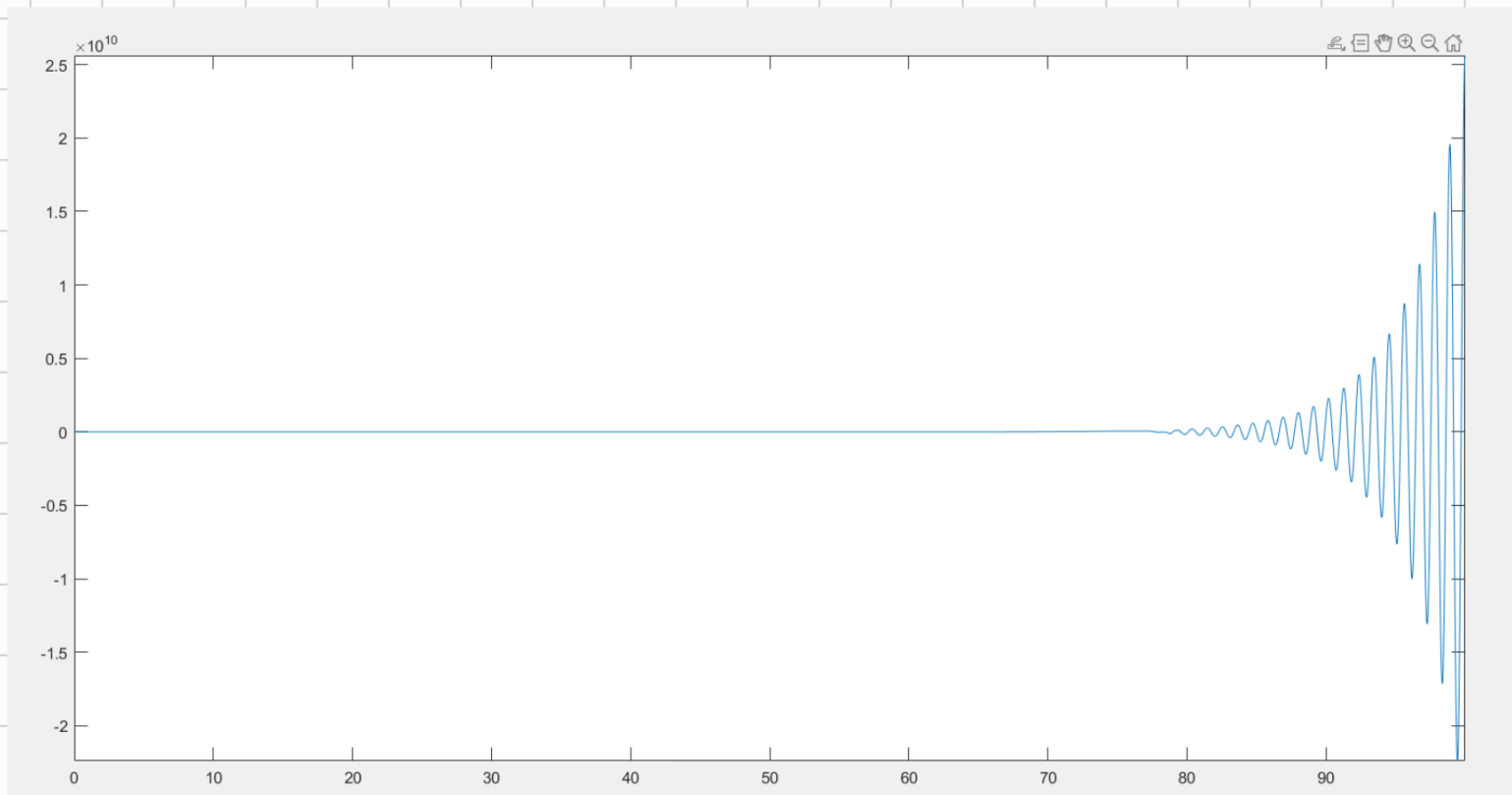
$$\Theta_1(s) = \frac{\frac{s^3}{1000} + \frac{303}{10000}s^2 + 3.3081s + \frac{921}{1000}}{s^4 + \frac{3}{50}s^3 + 66s^2 + \frac{54}{25}s + 1098}$$

$$\begin{aligned} &= \frac{-0.0107 - 0.2478i}{s - (-0.2467 + 5.7654i)} + \frac{-0.0107 + 0.2478i}{s - (-0.2467 - 5.7654i)} \\ &+ \frac{0.0157 + 0.2453i}{s - (-0.2767 + 5.7355i)} + \frac{0.0157 - 0.2453i}{s - (-0.2767 - 5.7355i)} \end{aligned}$$

A P₁ P₂
B

$$\theta_1(t) = Ae^{p_1 t} + \bar{A}e^{\bar{p}_1 t} + Be^{p_2 t} + \bar{B}e^{\bar{p}_2 t}$$

$$= 2e^{0.2467t} [-0.0107 \cos(5.7654t) - 0.2478 \sin(5.7654t)] \\ + 2e^{-0.2767t} [0.0157 \cos(5.7355t) - 0.2453 \sin(5.7355t)]$$



$$(3) \quad \frac{\theta_1(s)}{F} = \frac{0.025s + 2}{\frac{1}{9}s^4 + \frac{1}{180}s^3 + \frac{22}{3}s^2 + \frac{1}{5}s + 120}$$

$$\text{unit impulse} \Rightarrow F(s) = 1$$

$$\theta_1(s) = \frac{0.025s + 2}{\frac{1}{9}s^4 + \frac{1}{180}s^3 + \frac{22}{3}s^2 + \frac{1}{5}s + 120}$$

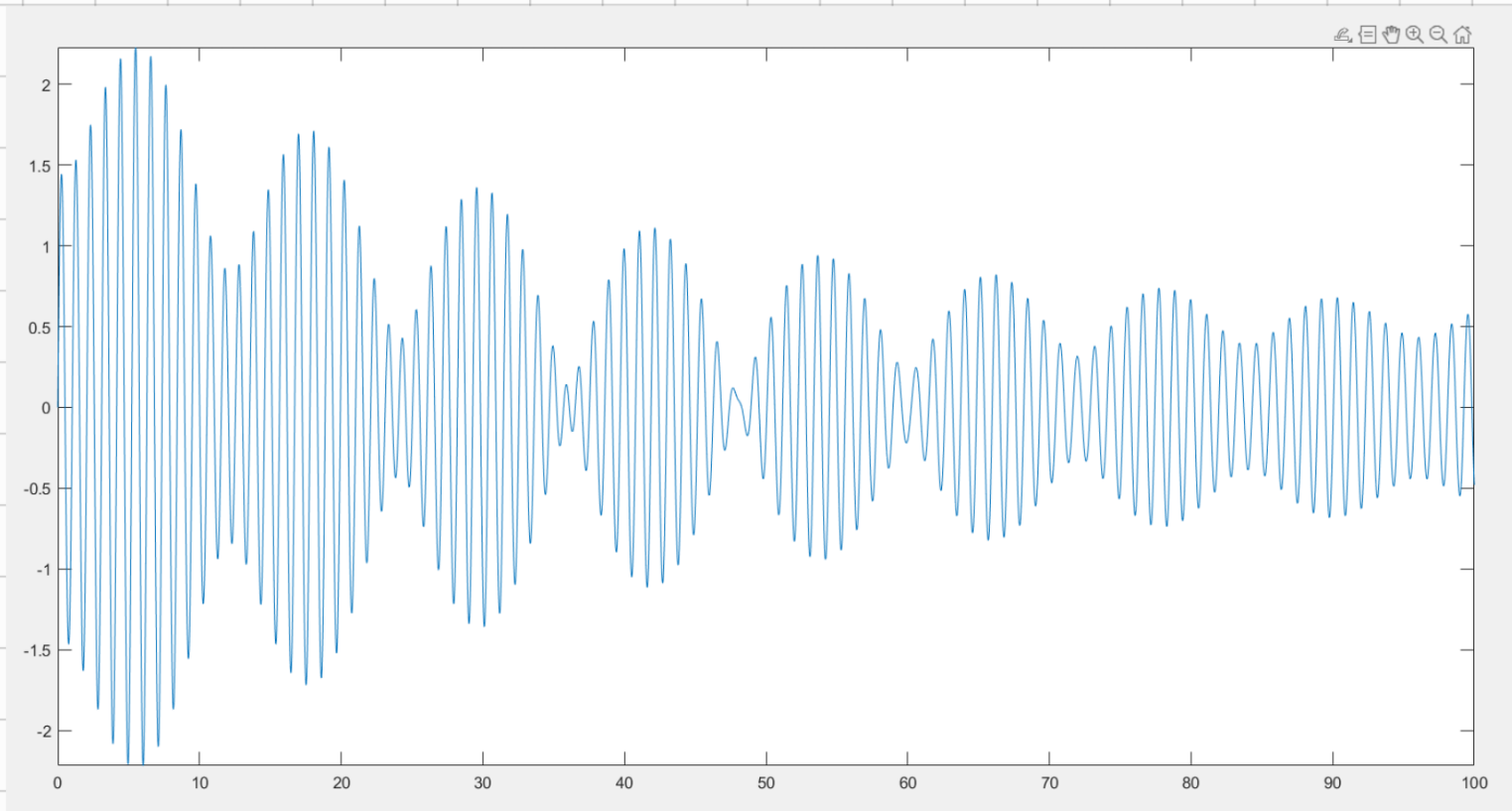
$$\text{zeros: } z = -100$$

$$\text{poles: } p_{1,2} = -0.03 \pm 5.999i \quad p_{3,4} = \pm 5.4772i$$

$$\Theta_1(s) = \frac{A \cdot 0.25i}{s - (-0.03 + 6i)} + \frac{-0.25i}{s - (-0.03 - 6i)} + \frac{B \cdot -0.2739i}{s - (5.4772i)} + \frac{0.2739i}{s - (-5.4772i)}$$

P_1
 P_2

$$\Theta_1(t) = 2e^{-0.03t} (-0.25 \sin 6t) + 2[-0.2739 \sin(5.4772t)]$$



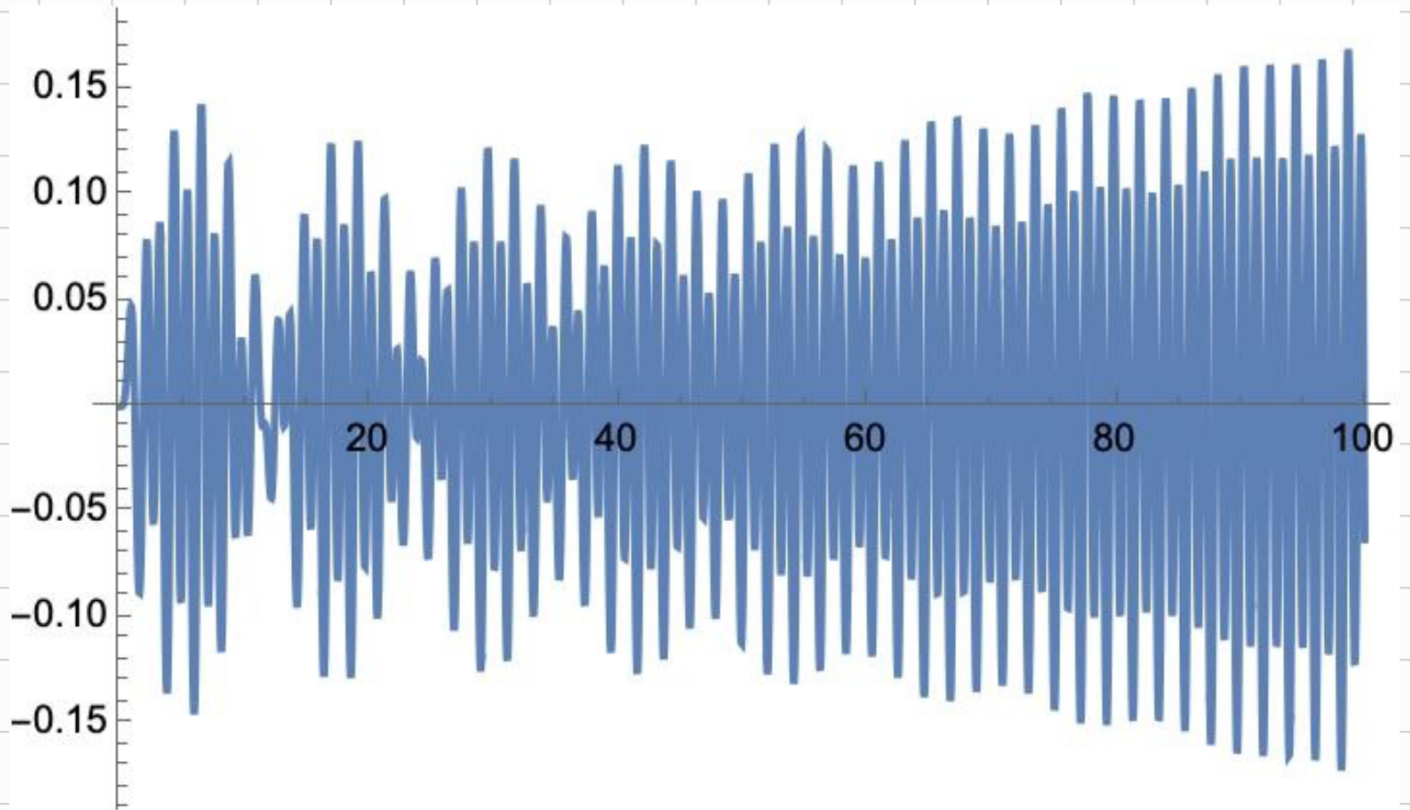
(4) $\frac{\Theta_1(s)}{\frac{3}{s^2+9}} = \frac{0.025s + 2}{\frac{1}{9}s^6 + \frac{1}{180}s^5 + \frac{25}{3}s^4 + \frac{1}{5}s^3 + 120s^2 + \frac{9}{5}s + 108}$ $F(s) = \frac{3}{s^2+9}$

$$\Theta_1(s) = \frac{0.06s + 6}{\frac{1}{9}s^6 + \frac{1}{180}s^5 + \frac{25}{3}s^4 + \frac{13}{50}s^3 + 18s^2 + \frac{9}{5}s + 108}$$

$$\Theta_1(s) = \frac{A \cdot (-0.0004 - 0.0278i)}{s - (-0.03 + 6i)} + \frac{-0.0004 + 0.0278i}{s - (-0.03 - 6i)} + \frac{B \cdot 0.0391i}{s - 5.4772i} + \frac{-0.0391i}{s + 5.4772i} + \frac{C \cdot 0.0004 - 0.0159i}{s - 3i} + \frac{0.0004 + 0.0159i}{s + 3i}$$

P_1
 P_2
 P_3

$$\begin{aligned} \theta(t) &= Ae^{P_1 t} + \bar{A}e^{\bar{P}_1 t} + Be^{P_2 t} + \bar{B}e^{\bar{P}_2 t} + Ce^{P_3 t} + \bar{C}e^{\bar{P}_3 t} \\ &= 2e^{-0.03t} \left[-0.0004 \cos 6t - 0.0278 \sin 6t \right] + 2 \left[-0.0391 \sin(5.4772t) \right] \\ &\quad + 2 \left[0.0004 \cos 3t - 0.0159 \sin 3t \right] \end{aligned}$$



$$15) D(s) = s^4 + \frac{3}{50}s^3 + 66s^2 + \frac{9}{5}s + 1080$$

$$s^4 \quad 1 \quad 66 \quad 1080$$

$$s^3 \quad \frac{3}{50} \quad \frac{9}{5}$$

$$s^2 \quad 36 \quad 1080$$

$$s^1 \quad 0(\varepsilon)$$

$$s^0 \quad 1080$$

$\Rightarrow$ no positive roots but have roots on image axis,
marginally stable